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Optimization of cellular automata-based routing using discrete Boltzmann model for urban pedestrian evacuation during flooding

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ABSTRACT

Flooding is one of the most severe problems faced in urban disaster issues today, threatening residents' lives and safety while causing significant property losses. Therefore, an effective route planning strategy is urgent to evacuate people to safe areas during flood events. In this study, a rainfall model using the multi-speed discrete Boltzmann model (DBM) is proposed in conjunction with a cellular automaton-based emergency evacuation and risk avoidance model. Using a suburban area as a case study, the model simulates the evacuation of people during actual floods. The spatial synchronization of flood inundation and human movement is achieved, using an extended Moore neighborhood framework, taking into account the critical preemptive response time during floods that influences evacuation and retreat processes. By comparing it with the traditional A* algorithm (a widely used heuristic shortest-path search method), the advantages and practicality of the proposed model are demonstrated. From a microscopic perspective, incorporating appropriate crowd behavior rules reveals rational evacuation behavior patterns during flooding. The results provide reference and guidance for urban flood control, disaster reduction, and emergency evacuation planning.

1. Introduction

In recent years, under the influence of both global climate change and rapid urbanization, significant changes have been occurring in the water cycle process and urban underlying surface elements. Extreme climate events are becoming more frequent and intense. The urban "heat island effect" and "rain island effect" have become more pronounced, altering runoff generation and concentration mechanisms. As a result, urban flood disasters have been increasingly exacerbated (Zhang et al., 2016; Song et al., 2013; Li et al., 2017; Xu and Cheng, 2019, 2020). Statistical data from the "China Flood and Drought Disaster Bulletin: 2018″ (Ministry of Water Resources of the People's Republic of China, 2018) show that, from 2000 to 2018, floods in China caused 21,720 deaths and direct economic losses of 3,163.952 billion yuan. Beginning in April 2024, unusual rainfall affected the United Arab Emirates and northern Oman, leading to at least 20 fatalities in Oman and 4 in the United Arab Emirates. The severe rainfall caused significant damage to infrastructure and disrupted the livelihoods of people in the affected areas (Fraser-Baxter, 2024).

Therefore, the importance of people-oriented and non-engineering flood control measures has gradually received the attention of governments around the world. At the Third World Conference on Disaster Risk Reduction in 2015, "strengthening disaster preparedness to improve emergency response capabilities" was listed as one of the four priority actions, emphasizing the need to "enhance the capacity of local governments to evacuate residents in vulnerable areas" (Yang et al., 2023). China has put forward the requirement in the "National Medium- and Long-Term Science and Technology Development Plan Outline (2006–2020)" to "focus on researching and developing key technologies for monitoring, early warning, and emergency response to earthquakes, typhoons, rainstorms, floods, geological disasters, etc." (State Council of the People's Republic of China, 2005). In the "14th Five Year Plan for the

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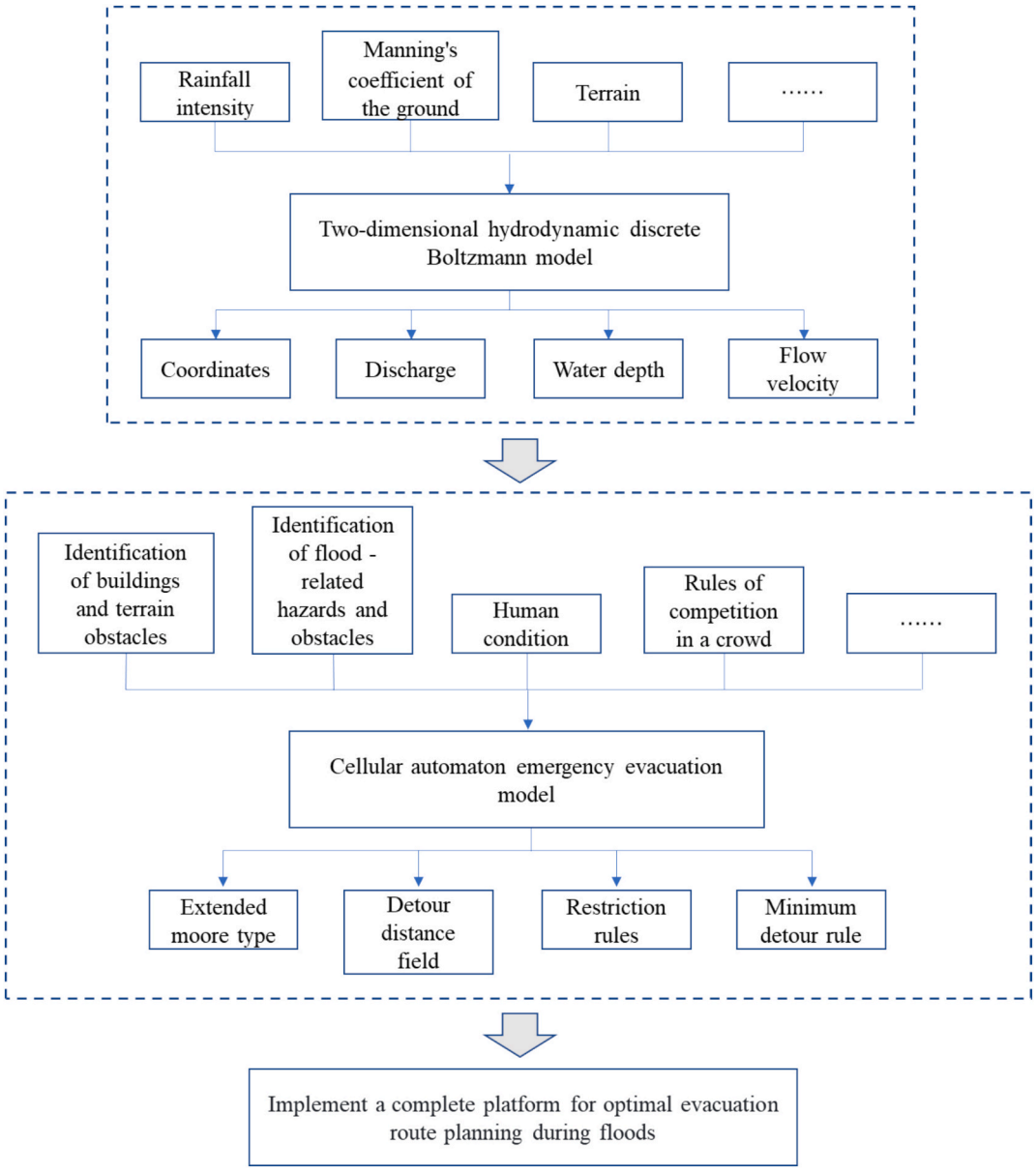


Fig. 1. Technical flowchart of the model research.

National Emergency Management System" issued in 2021, it was particularly emphasized to "improve the planning and layout of emergency shelters and build comprehensive emergency shelters in key urban agglomerations" (State Council of the People's Republic of China, 2021).

Formulating a reasonable emergency evacuation plan in flooding is one of the key measures to protect lives and property. With the advancements in computing power, computer simulation has become one the major approaches (Capart, 2013; Ge et al., 2022; Zhang et al., 2024a; 2024b). Burstedde et al. (2001) were the first to apply a two- dimensional cellular automaton model to simulate the pedestrian dynamic and established a basic evacuation model. The optimal evacuation route plan has been a key concern in recent years (Trindade et al., 2018; Park et al., 2020; Yang et al., 2025). The Dijkstra algorithm is a classic shortest path algorithm (Marchese et al.,2002; Zhang et al., 2004). Its advantage lies in ensuring the finding of the shortest path while maintaining good robustness (Jasika et al., 2012). However, since this algorithm requires traversing all nodes, it leads to a lot of unnecessary calculations. Thus, heuristic search algorithms emerged for improving efficiency. A representative of them is the so-called A* algorithm (Hart and Nilsson, 1968). However, considering the actual distribution of obstacles, and due to the limitations of neighborhood search, the A* algorithm may lead to detours in the path and may fail in finding the global shortest path (Peng, 2018). Although some researchers have improved and optimized the above two algorithms (Liu and Zhao, 2009; Dong and Huang, 2012; Li et al., 2012; Knyazkov et al., 2014; Ren et al., 2015), improvements are still needed in terms of global optimality and applicability.

Currently, the shortest path algorithms mainly conduct the evaluation based on two-dimensional planar distance and rarely consider the impact of terrain undulation on path quality. Therefore, the applicability of these algorithms in different areas might be in question, especially in areas with significant terrain elevation changes (Sun et al., 2014; Zhao et al., 2017). Thus, it is necessary to develop an optimal evacuation path that is applicable to both flat and mountainous areas.

In addition, determining the best evacuation route is also affected by many other factors, such as flood response perception (Soon et al., 2018), flood warning (Parker and Priest, 2012; Nieland and Mushtaq,

2016), emergency resource reserves (Maura, 2016; Zhu et al., 2024a; 2024b), and individual human behaviors, age, gender, psychology, education, and physical condition, all of them can affect the formulation of flood avoidance strategies (Zheng and Cheng, 2011; Aerts et al., 2018; Bernardini et al., 2020). Bernardini et al (2017) proposed a method to estimate the moving speed of evacuees by using the results of hydrodynamic evolution as input. However, this model did not fully consider the slope changes around the evacuation area and the impact of flood dynamics on the evacuation of the crowd. Subsequently, Du et al. (2017) discussed how social media dynamically affects people's awareness of flood hazards and the subsequent evacuation process. In addition, people usually use water flow velocity and water depth as indicators to evaluate whether an area is a dangerous area (Chen et al., 2019; Zhu et al., 2024b; Liu et al., 2024). These factors can help decision makers plan evacuation strategies. Therefore, there is an urgent need to optimize the evacuation routes of the crowd in a continuous and dynamic manner for constructing a practical and effective flood emergency evacuation path optimization model.

In this work, the multi − speed discrete Boltzmann model (DBM) (LaRocca et al., 2015; Meng et al., 2018, Peng et al., 2023 and 2024) with a rainfall source term will be used to simulate the flooding caused by rainstorms, and the cellular automaton evacuation model will be used to simulate the evacuation process of people. In the cellular automaton evacuation model, the dynamic processes of terrain environment, flood, and people, the prediction time of people for floods, and the individual conditions of people will be considered, so that the established flood emergency evacuation model can more completely simulate the actual evacuation process.

This study presents three key innovations:

Coupled flood–evacuation modeling: A hybrid framework integrating the DBM and CA is proposed. It achieves spatial–temporal synchronization of flood inundation and pedestrian movement, overcoming the traditional separation between the two processes.
Novel dynamic route optimization mechanism: The CA-ERO model employs an extended Moore neighborhood for global search and integrates key flood-related factors—terrain, depth, velocity, and obstacles—into path updates, outperforming traditional methods such as A* in complex flood environments.
Comprehensive behavioral and performance integration: By jointly modeling response time, strength, and panic, the framework captures behavioral diversity in evacuation. A multi-criteria evaluation and sensitivity analysis further validate its enhanced performance compared with existing algorithms.

2. Model framework

This model is mainly composed of two parts, a discrete Boltzmann model for two-dimensional shallow water flows and a cellular automaton-based emergency evacuation model for pedestrians. The model framework is illustrated in Fig. 1.

First, based on the known rainfall information data and relevant materials, the discrete Boltzmann model for two-dimensional shallow water flows is utilized to simulate the flooding induced the rainfall. Second, the impact of floods on human evacuation behavior is analyzed. The judgment of flood related obstacles is set up. Additionally, the identification of building and slope-related obstacles, the state of people, and competition rules are incorporated. Then, the cellular automaton-based emergency evacuation model is adopted to obtain the shortest evacuation route. Ultimately, an optimal flood emergency evacuation model that is more in line with the practical situation is achieved.

2.1. 2D shallow water equation solution using discrete Boltzmann method

In order to obtain hydrodynamic fields, the 2D shallow water equations (SWEs) are used, which can be written as follows,

$$\frac{\partial h}{\partial t}+\nabla\cdot(h\boldsymbol{u})=R, \tag{1}$$

$$\frac{\partial(h\boldsymbol{u})}{\partial t}+\nabla\cdot(h\boldsymbol{u}\boldsymbol{u})=-\nabla\left(\frac{gh^2}{2}\right)+\nu\nabla\cdot\nabla(h\boldsymbol{u})+\boldsymbol{F}, \tag{2}$$

where h represents the water depth, g the gravitational acceleration (9.81 m/s^2), $\boldsymbol{u}$

the depth-averaged water velocity , t the time , and$R$ the rainfall intensity (Ding et al., 2018). $\boldsymbol{F}$ represents the source term given by

$$\boldsymbol{F}=-gh\frac{\partial z_b}{\partial \boldsymbol{x}}-\frac{\boldsymbol{\tau_b}}{\rho}, \tag{3}$$

where $\boldsymbol{\tau_b}$ is defined as

$$\boldsymbol{\tau_b}=\rho C_b\boldsymbol{u}\sqrt{\boldsymbol{u}\cdot\boldsymbol{u}}, \tag{4}$$

In the above equation, C_b is the dimensionless bed friction coefficient, which can be calculated as:

$$C_b=\frac{g}{C_z^2}, \tag{5}$$

where C_z is the bed friction coefficient and can be calculated using the following formula:

$$C_z=\frac{h^{\frac{1}{6}}}{n_b}, \tag{6}$$

When using the DBM model, it is often necessary to nondimensionalize the physical quantities using the following expressions,

$$\widehat{\boldsymbol{r}}=\frac{\boldsymbol{r}}{h_0},\ \widehat{\boldsymbol{V}}=\frac{\sqrt{2}\boldsymbol{V}}{\sqrt{gh_0}},\ \widehat{h}=\frac{h}{h_0},\ \widehat{\boldsymbol{F}}=\frac{2\boldsymbol{F}}{g}, \tag{7}$$

$$\widehat{c}=\frac{\sqrt{2}c}{\sqrt{gh_0}},\ \widehat{t}=\frac{\sqrt{gh_0}}{\sqrt{2}h_0},\ \widehat{f}=f\frac{g}{2},\ \widehat{\nu}=\frac{\nu}{\nu_0},\ \widehat{P}=\frac{P}{P_0}=\widehat{h}^2, \tag{8}$$

After nondimensionalization and removing the hat symbol, the discrete Boltzmann equation is written as,

$$\frac{\partial f_\alpha}{\partial t}+\boldsymbol{c}_\alpha\cdot\frac{\partial f_\alpha}{\partial \boldsymbol{x}}=-\frac{1}{\tau}\left(f_\alpha-f_\alpha^{eq}\right)+\mathscr{F}_\alpha \tag{9}$$

which describes the evolution of the mesoscopic discrete distribution function $f_\alpha(\boldsymbol{x},t)$ for the discrete velocity $\boldsymbol{c}_\alpha$ at a given position $\boldsymbol{x}$ and time t. Accordingly, water depth h and velocity $\boldsymbol{u}$ are calculated as:

$$h=\sum_\alpha f_\alpha=\sum\nolimits_\alpha f_\alpha^{eq}, \tag{10}$$

$$\boldsymbol{u}=\frac{1}{h}\sum\nolimits_\alpha f_\alpha\boldsymbol{c}_\alpha=\frac{1}{h}\sum\nolimits_\alpha f_\alpha^{eq}\boldsymbol{c}_\alpha, \tag{11}$$

The mesoscopic body force term $\mathscr{F}_\alpha$ is calculated as

$$\mathscr{F}_\alpha=h\omega_\alpha[\boldsymbol{F}(\boldsymbol{x},t)\cdot\boldsymbol{c}_\alpha]+\omega_\alpha R. \tag{12}$$

The dimensionless relaxation time τ can be refined by $\tau=\widehat{h}/\kappa$ and the expression for κ is

$$\kappa=\frac{\nu_0\sqrt{gh_0/2}}{P_0}=\frac{\nu_0\sqrt{gh_0/2}}{gh_0^2/2}. \tag{13}$$

In order to simulate supercritical flow, the following fourth-order form of equilibrium distribution function is adopted (Meng et al., 2018, Peng et al. 2023 and 2024), i.e.,

Table 1
Discrete velocities and weights of D2Q16 set.

Roots (c_α)	Weights (ω_α)
−2.3344142183389773	0.04587585476806826
2.3344142183389773	0.04587585476806826
0.7419637843027259	0.45412414523193156
−0.7419637843027259	0.45412414523193156

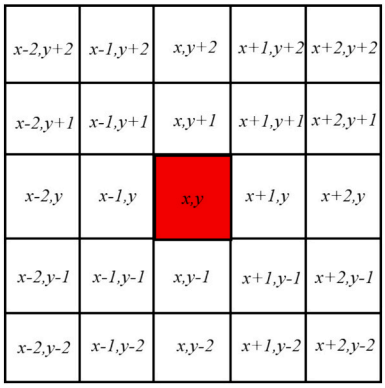


Fig. 2. Extended Moore neighborhood.

Table 2
Relationship between the safety level of floodwaters, water depth and flow velocity.

Water depth(m)	$h < 0.3$	$0.3 < h < 0.6$	$h > 0.6$
Flow velocity(m/s)			
$v < 0.4$	safe	safe	dangerous
$0.4 < v < 0.8$	safe	dangerous	extremely dangerous
$v > 0.8$	dangerous	extremely dangerous	extremely dangerous

$$f_\alpha^{eq} = h\omega_\alpha\{1+\boldsymbol{c}_\alpha \bullet \boldsymbol{u}+\frac{1}{2}\left[(\boldsymbol{c}_\alpha \bullet \boldsymbol{u})^2 - \boldsymbol{u}\bullet\boldsymbol{u} + (h-1)(\boldsymbol{c}_\alpha \bullet \boldsymbol{c}_\alpha - 2)\right] + \frac{\boldsymbol{c}_\alpha \bullet \boldsymbol{u}}{6}\left[(\boldsymbol{c}_\alpha \bullet \boldsymbol{u})^2 - 3\boldsymbol{u}\bullet\boldsymbol{u} + (h-1)(\boldsymbol{c}_\alpha \bullet \boldsymbol{c}_\alpha - 4)\right] + \frac{1}{24}\left[(\boldsymbol{c}_\alpha \bullet \boldsymbol{u})^4 - 6\boldsymbol{u}\bullet\boldsymbol{u} + (h-1)(\boldsymbol{c}_\alpha \bullet \boldsymbol{c}_\alpha - 4)\right] + \frac{h-1}{4}\left[(\boldsymbol{c}_\alpha \bullet \boldsymbol{c}_\alpha - 4)\left((\boldsymbol{c}_\alpha \bullet \boldsymbol{u})^2 - \boldsymbol{u}\bullet\boldsymbol{u}\right) - 2(\boldsymbol{c}_\alpha \bullet \boldsymbol{u})^2\right] + \frac{(h-1)^2}{8}\left[(\boldsymbol{c}_\alpha \bullet \boldsymbol{c}_\alpha)^2 - 8\boldsymbol{c}_\alpha \bullet \boldsymbol{c}_\alpha + 8\right]\}. \quad (14)$$

Together with this fourth-order equilibrium distribution function, we use a multispeed D2Q16 discrete velocity set, which is listed in Table 1.

In order to solve Equation (10), a finite difference method can be chosen to discretize both time and space. In this work, an upwind scheme is used for spatial discretization, and a forward Euler scheme is used for time discretization.

2.2. Cellular automaton-based emergency evacuation route planning model

With the continuous advancement of remote sensing and geographic information system (GIS) technologies, the acquisition of high-resolution spatial information has become more feasible. Currently, researchers are increasingly utilizing cellular automata (CA) models and leveraging their powerful data analysis and modeling capabilities to simulate the behavior and evolution of urban systems. The approach overcomes limitations of using equations and parameters since it is easy to implement and is capable of simulating complex phenomena. Generally, the state set of a cellular automaton can be represented as:

$$A = f(L, d, S, N), \quad (15)$$

where L represents the cellular grid space, d the spatial dimension; S the finite discrete state set (e.g., 0, 1),N the possibility of all cells in a given neighborhood, and f the local mapping or local rule. The state of each cell at the next time step is determined by its current state ‘and the state of all neighboring cells. At each time step, all cell states are updated simultaneously. The evolution could be written in a general form

$$S^{t+1} = f(S^t, N), \quad (16)$$

where S^t represents the set of cell states at time t, N the set of neighboring states, f the state transition function, and S^{t+1} the set of cell states at time$t+1$. f is typically constructed using rules and is independent of time and the cell itself.

The route optimization algorithm in this study is based on the geographic cellular automata algorithm utilizing an extended Moore neighborhood (Shi et al., 2016). There are 24 possible movement directions for individuals, with the central cell coordinates being (x, y), and the neighbor indices are shown in Fig. 2. The following rules are established to constrain the movement to outer cells: When (x-1, y + 1) is an obstacle, movement to (x-2, y + 2), (x-2, y + 1), and (x-1, y + 2) is not allowed; when (x-1, y) is an obstacle, movement to (x-2, y), (x-2, y + 1), and (x-2, y-1) is not allowed. This logic extends similarly for other cases. Additionally, when multiple cells in the inner layer are obstacles, the constraints on the outer cells can be cumulatively applied.

Cell states can be divided into three types: accessible cells ($S = 0$), people ($S = 1$), and obstacles ($S = -1$). The determination principle is as follows:

$$S = \begin{cases} -1, & \textit{hazardous points, buildings, or significant elevation variations} \\ 1, & \textit{cells occupied by people} \\ 0, & \textit{otherwise} \end{cases}, \quad (17)$$

In this case, the obstacles generally include two categories: the first is danger zones and buildings caused by flooding, and the second category is the grid unit with significant elevation variations in adjacent terrain, which are typically judged using the following equation:

$$\text{If}\Delta h = |h_1 - h_2| < P, \quad (18)$$

where h_1 and h_2 are the elevation values of two adjacent cells, and P is the obstacle-crossing ability considered based on human behavior characteristics. In this study, P is set to 1.2 m. If the elevation difference Δh is greater than P, the adjacent cell is considered a large elevation difference obstacle. Conversely, if the difference is less than P, the cell is considered accessible. This judgment ensures that individuals avoid entering excessively steep locations during evacuation, thereby ensuring their safety. Any cell that satisfies one of the two conditions should be identified as an obstacle.

The identification of flooding hazard obstacles is also a key component. Liu et al. (2018) and Wang et al. (2019) summarized domestic and international research on flood hazard, focusing on classification and grading methods for evaluating the danger levels of different populations and flood-prone areas. Bernardini et al. (2020) experimentally quantified human motion speeds at different flood depths to support evacuation modeling. They concluded with the following Table 2 showing the relationship between people and flood depth, and flow velocity.

Meanwhile, we also refer to the formula for the initial velocity of human body slipping and toppling instability in floods proposed by Xia et al. (2014) to further refine the formulation of obstacle judgment, and the formula is

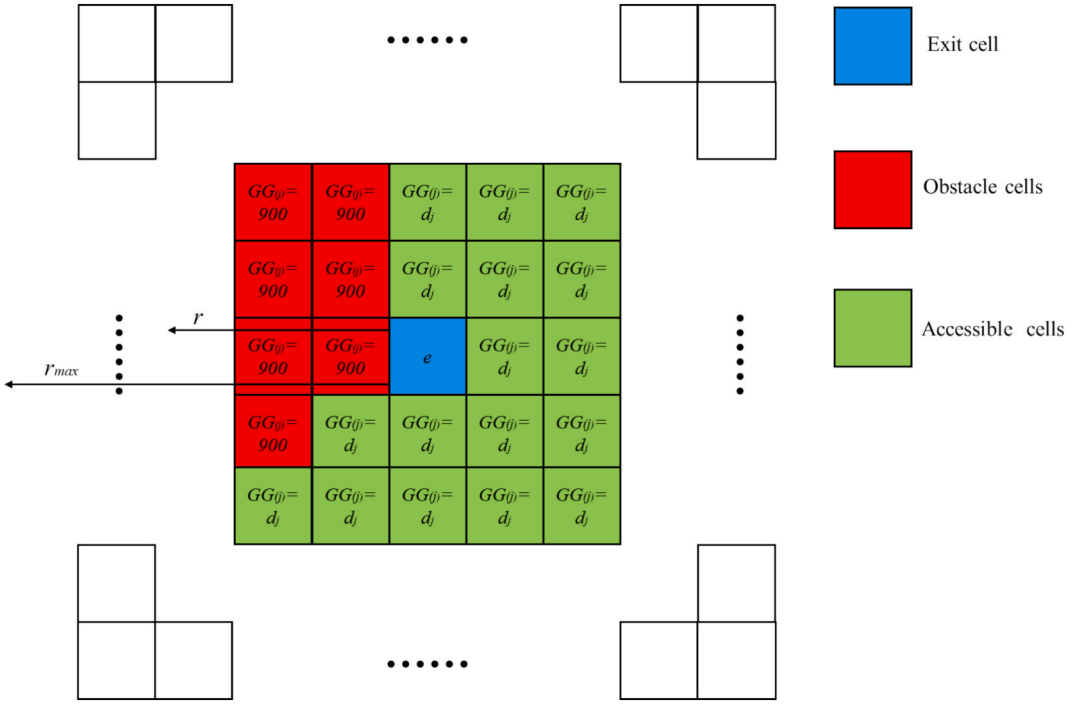


Fig. 3. Traversal schematic diagram.

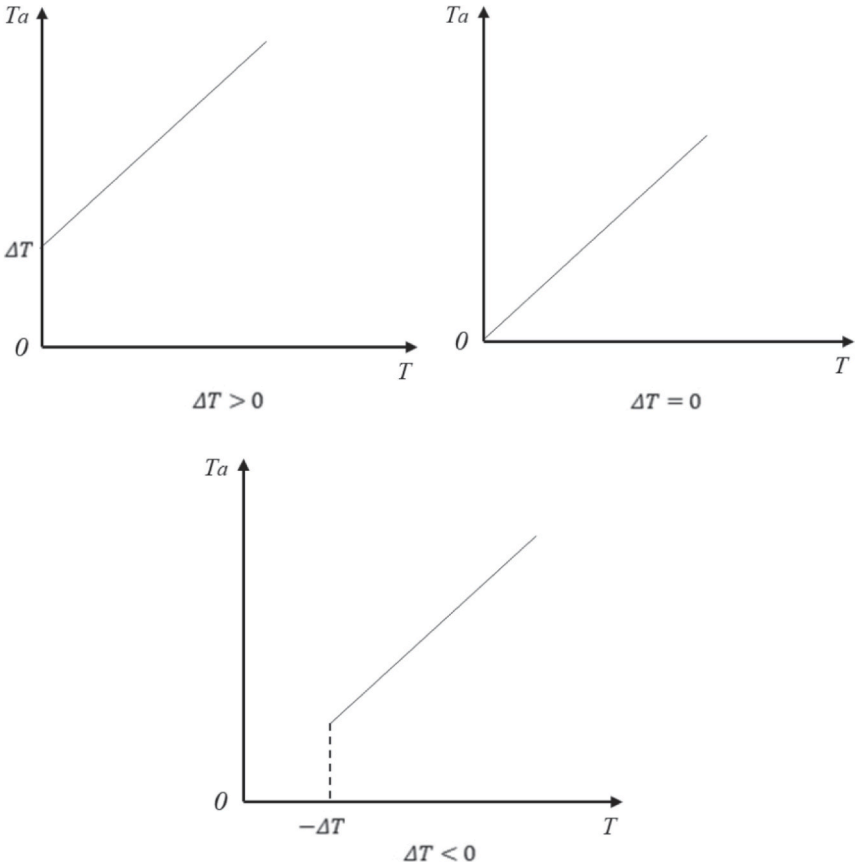


Fig. 4. Variation of T_a against T with different ART.

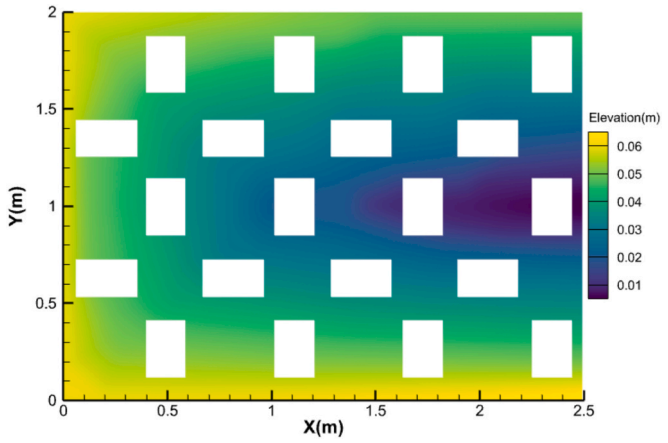


Fig. 5. Rainfall-runoff experiment topographic layout map.

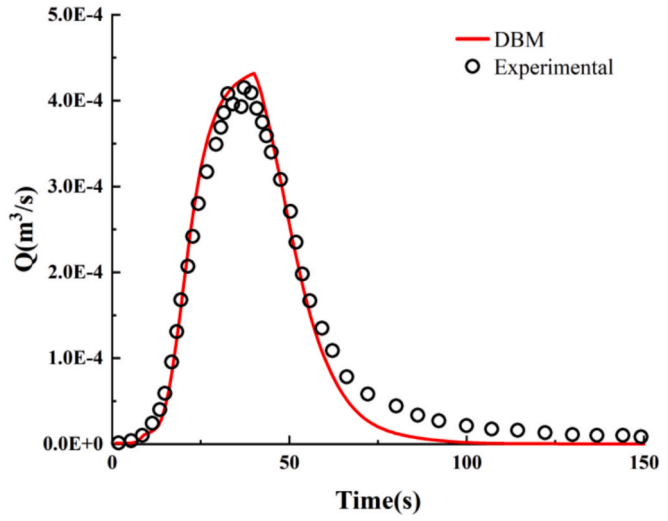


Fig. 6. Comparison between simulated results and experimental data at the domain outlet for rainfall.

$$V_c = \alpha\left(\frac{h_f}{h_p}\right)^{\beta}\sqrt{\frac{m_p}{\rho_f h_f^2} - \left(\frac{a_1}{h_p^2} + \frac{b_1}{h_f h_p}\right)(a_2 m_p + b_2)}, \tag{19}$$

where V_c is the critical velocity at which a person becomes unstable in floodwaters, ρ_f the density of water (1000 kg/m$^{3)}$, m_p the body weight of a person, and h_p the height of a person. In this study, based on the average physical attributes of Chinese adults (Guo et al. 1995), the calibrated parameter values are $a_1 = 0.633$, $b_1 = 0.367$, $a_2 = 0.001015$ m^3/kg, and $b_2 = -0.004927$ m^3, respectively. The determination of α and β is related to the shape, friction coefficient and drag coefficient of the object. Referring to the experimental results of Xia et al. (2014), the fitted values are $\alpha = 3.472$ m$^{0.5}$/s, $\beta = 0.188$.

Meanwhile, the combined influence of water depth and flow velocity also needs to be considered. According to the ratio of the incoming flow velocity to the critical velocity given by Martínez – Gomariz et al. (2016) and Guo (2017), the degree of human danger is defined as:

$$HD = \mathrm{Min}(1.0, V/V_c) \tag{20}$$

In the above formula, HD represents the degree of danger to the human body in floodwaters, and V is the average velocity along the

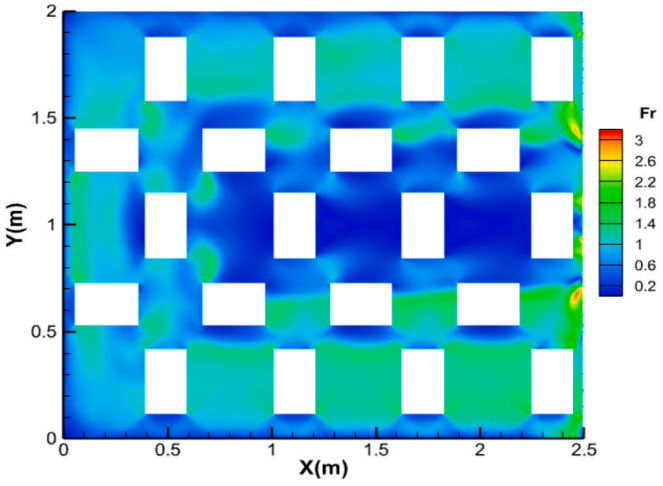


Fig. 8. Froude number distribution from DBM simulation at t = 40 s.

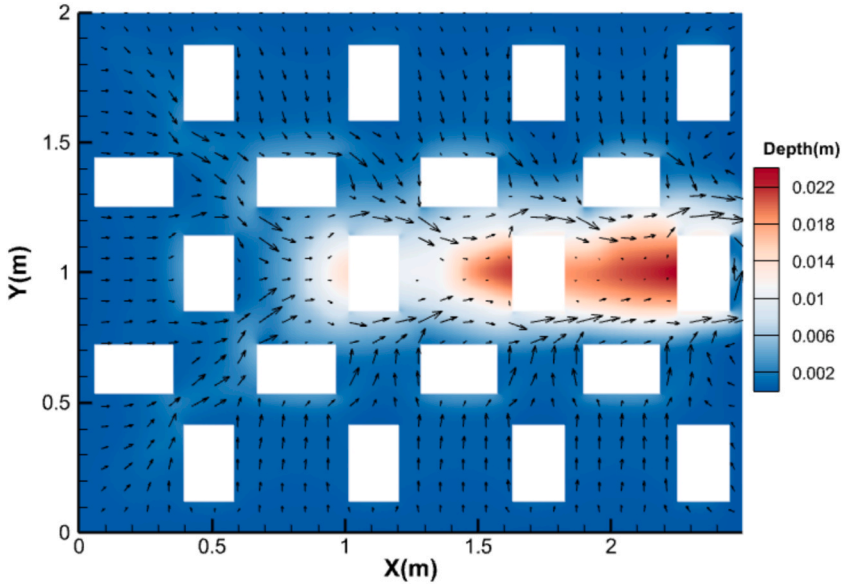


Fig. 7. Water depth and flow field from DBM simulation at t = 40 s.

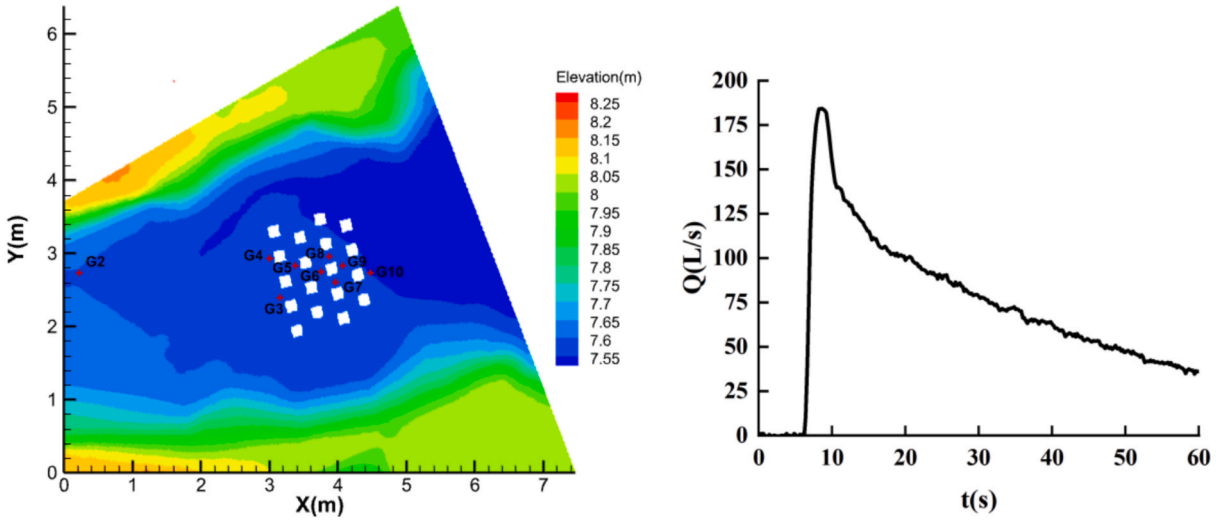
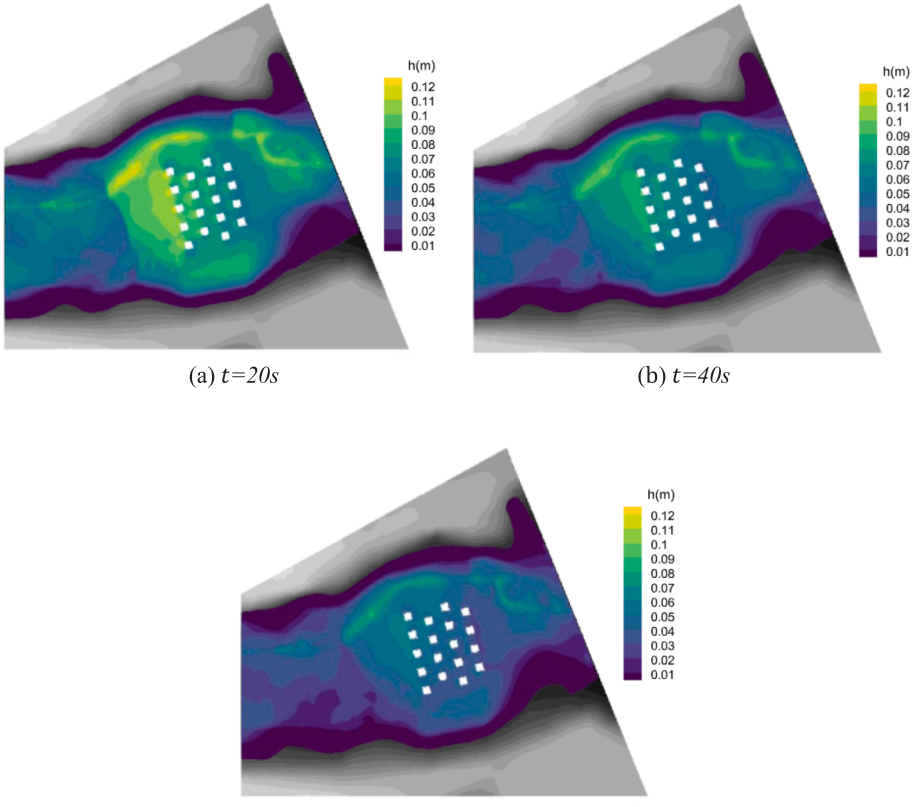


Fig. 9. (a) Configuration of the physical urban flood model; (b) inflow hydrograph applied at the upstream boundary.



(a) *t=20s* (b) *t=40s*

(c) *t=60s*

Fig. 10. Computed water depth distribution for the urban flash-flood case.

vertical direction at a certain moment. Equations (19) and (20) comprehensively consider the effects of the two key factors, water depth and flow velocity, on human body stability. In this way, the index *HD* can not only judge the degree of danger to people when the water depth is relatively large but is also applicable to the situation where the water depth is relatively shallow but the flow velocity is relatively high. Thus, it can ensure the safety of evacuees to a greater extent.

In summary, it is assumed that $h_p = 1.7\,\mathrm{m}$ and $m_p = 65\,\mathrm{kg}$ in this study. Based on the above − mentioned formulas, areas where $h > 0.6\,\mathrm{m}$ or $v > 0.8\ \mathrm{m/s}$ or $HD \geq 0.7$ are identified as dangerous zones for people and treated as obstacles.

Considering the distribution of global obstacles and flood related dangerous zones, the algorithm uses the exit position as the starting point. It calculates the shortest path distance from the exit to all reachable cells through distance superposition. A detour distance matrix G of the same size as the spatial discretization needs to be set up. $G(x, y)$ represents the detour distance value of the point (x, y). At the initial moment, the detour distance of the cell at the exit position is set as $G = 0$, meaning there is no detour, and the detour distance $G(x, y)$ of the remaining cells is set to a relatively large value of 900.

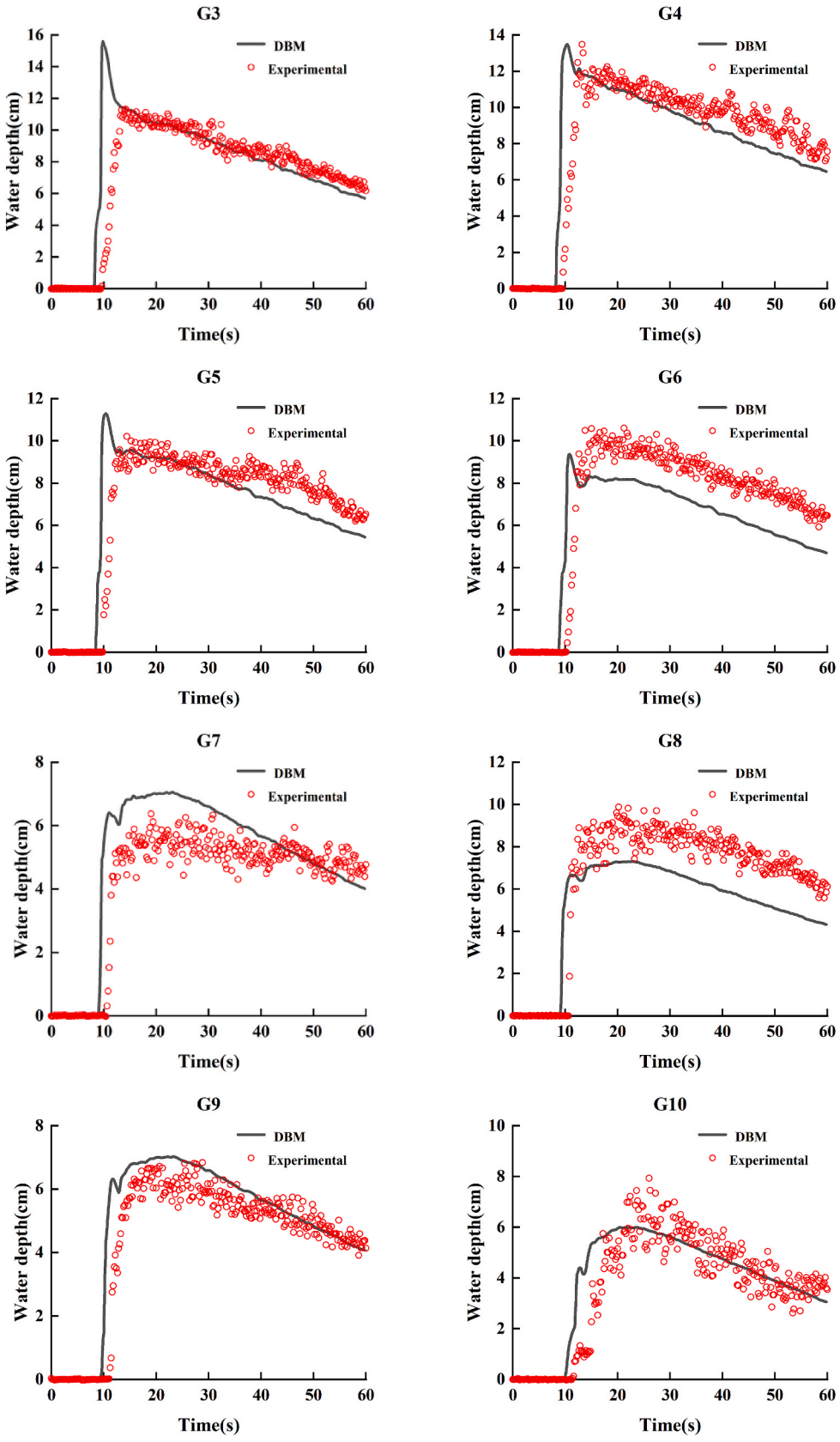


Fig. 11. Temporal evolution of water depth recorded at eight monitoring points within the urban flood scenario.

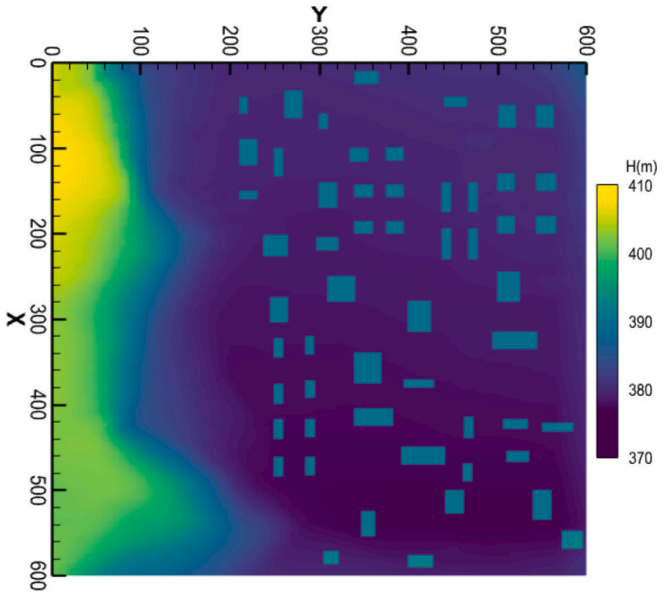


Fig. 12. Layout of the region.

Taking e as the current central cell, the superimposed distance from it to the neighboring cell is d_j, and its calculation formula is as follows:

$$d_j = \sqrt{((x_c - x_j)*dx)^2 + \left(\left(y_c - y_j\right)*dy\right)^2 + (z_c - z_j)^2}, \tag{21}$$

where z_c and z_j ($j \in \{1, 2, 3, \ldots, 24\}$) represent the elevation of the central cell c and the neighboring cell j, respectively; dx and dy are the length and width of a cell.

Based on the current position of the central cell, the status value $S(j)$ of the 24 neighboring cells is checked one by one. If the neighboring cell j has a state of obstacle, a temporary detour distance $GG_{(j)}$ is set to 900. If j is an accessible cell, the distance is calculated using the distance summation formula:

$$GG_{(j)} = G_c + d_j, \tag{22}$$

After the distance superposition calculation is completed, the following formula is used to make a judgment for all neighboring cells j.

$$G_{(j)} = min\left(G_{(j)}, GG_{(j)}\right), \tag{23}$$

which take the smaller of the $G_{(j)}$ value of its adjacent cell and the temporary detour distance $GG_{(j)}$ as the detour distance of the neighbor j.

The traversal order of the distance superposition algorithm for the entire cellular space is depicted in Fig. 3. These calculations start from the exit. After calculating the superposition distances of the 24 neighboring cells of the exit, each neighboring cell is then taken as the center to continue calculating the superposition distances. As shown in the figure, r represents the vertical or horizontal coordinate distance between the cell with the superposition distance and the central cell ($r = 0$, 1, 2,…, $rmax$), and $rmax$ represents the maximum radius that ensures the traversal of all accessible cells. This makes the calculation expand circularly outward from the center of the exit in sequence until all cells are updated, and finally, the detour distance matrix is obtained.

In real-life scenarios, escape route planning is a dynamic problem evolving with flooding. Flood-prone areas change over time, and people's evacuation routes also change dynamically in response. In this study, the criteria for evaluating the quality of a route are not only the shortest path but also sufficient safety considerations in terms of space and time. Therefore, the anticipated response time (ART) ΔT is introduced, which represents the difference between the current flood time T and the anticipated or predicted flood time T_a when people predict the flood will arrive. The ART could be divided into three cases, $\Delta T > 0$, $\Delta T = 0$, and $\Delta T < 0$. Following the linear change law of T and T_a assumed by He (2018), the variation of T_a against T is depicted in Fig. 4.

When $\Delta T > 0$, people could take evacuation actions in advance. When $\Delta T = 0$, people's response to the occurrence of a flood is synchronous, and they change their evacuation routes following the changes of the flood. The relationship between ΔT and T_a can be expressed by the formula

$$T_a = T + \Delta T, \tag{24}$$

When $\Delta T < 0$, it indicates that people's prediction of the flood occurrence lags behind the current flood evolution. During the period $0 < T < -\Delta T$, people are unaware of the approaching flood and will

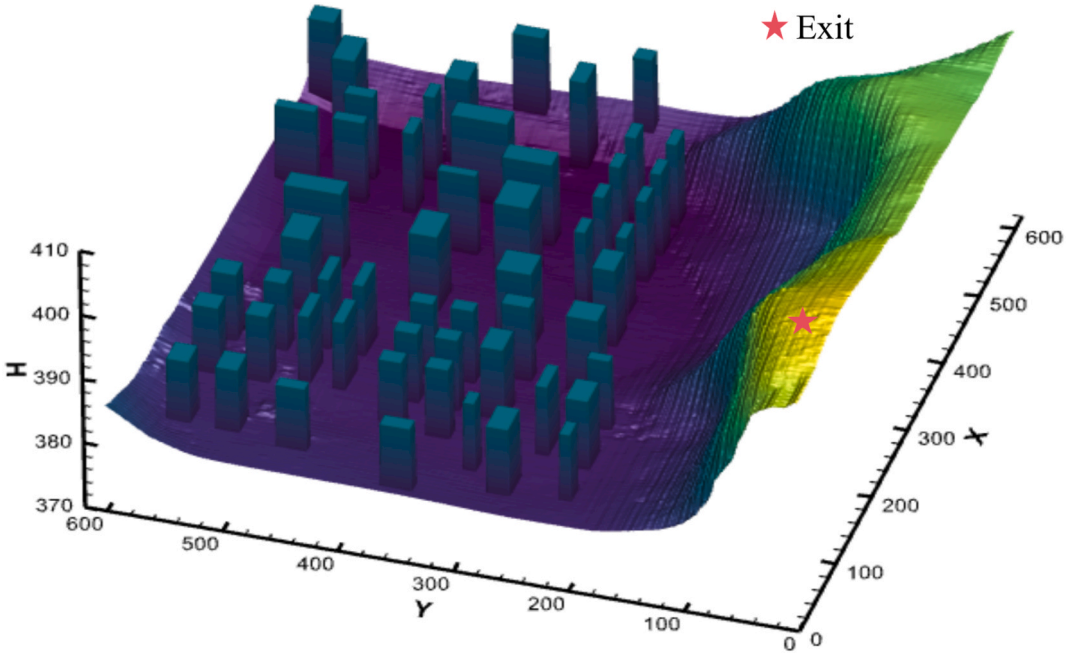
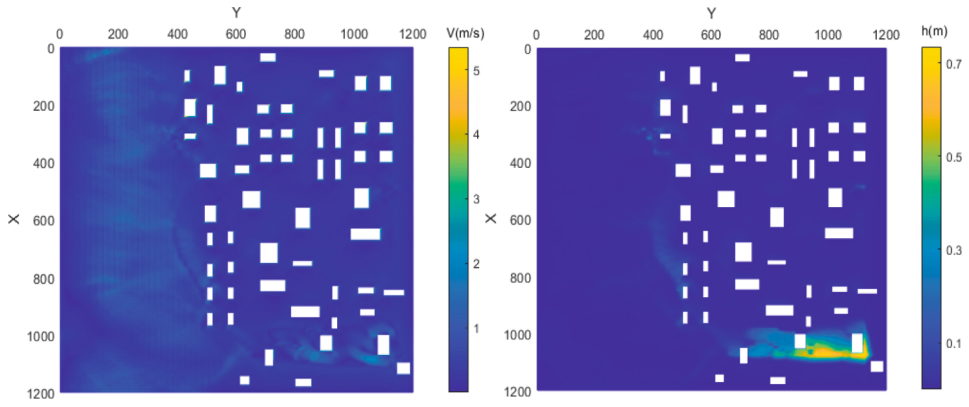
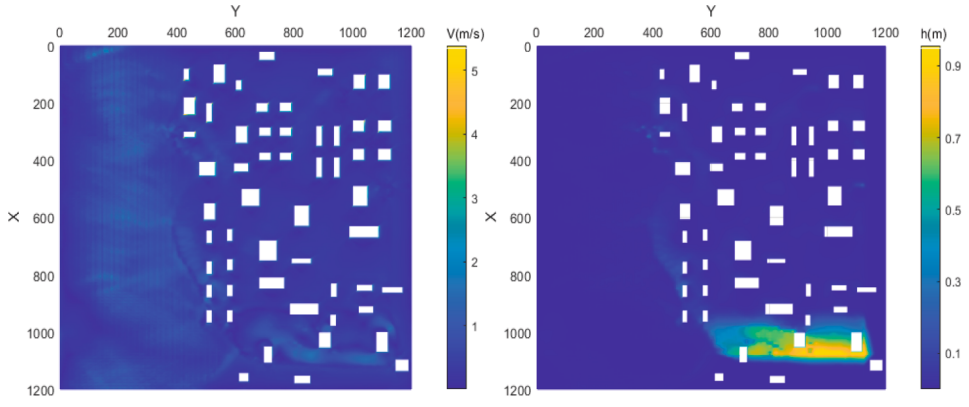


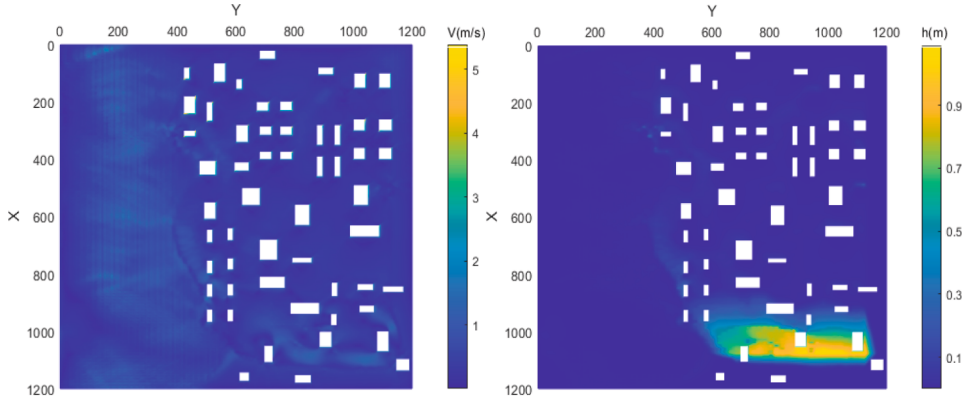
Fig. 13. 3D terrain map of the region.



(a) T=5 *min*



(b) T=8 *min*



(c) T=10 *min*

Fig. 14. Flood evolution process at different time points:(a) $T = 5$ min; (b) $T = 8$ min; (c) $T = 10$ min; (d) $T = 20$ min; (e) $T = 50$ min.

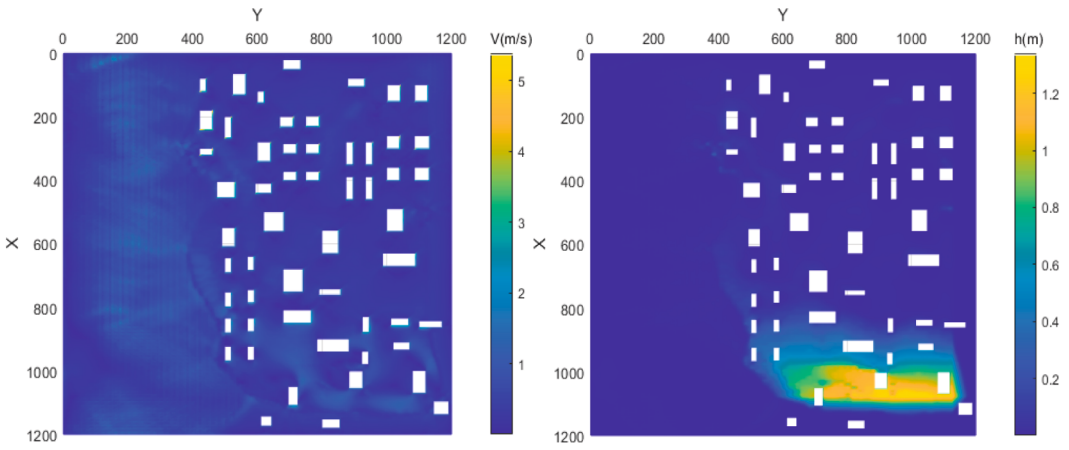
remain in place. Only when $T > -\Delta T$ will people make decisions based on the environmental information at the current time T. The relationship between ΔT and T_a is expressed by the following formula:

$$T_a = \begin{cases} \text{stagnation} & 0 < T \le -\Delta T \\ T & T > -\Delta T \end{cases}, \tag{25}$$

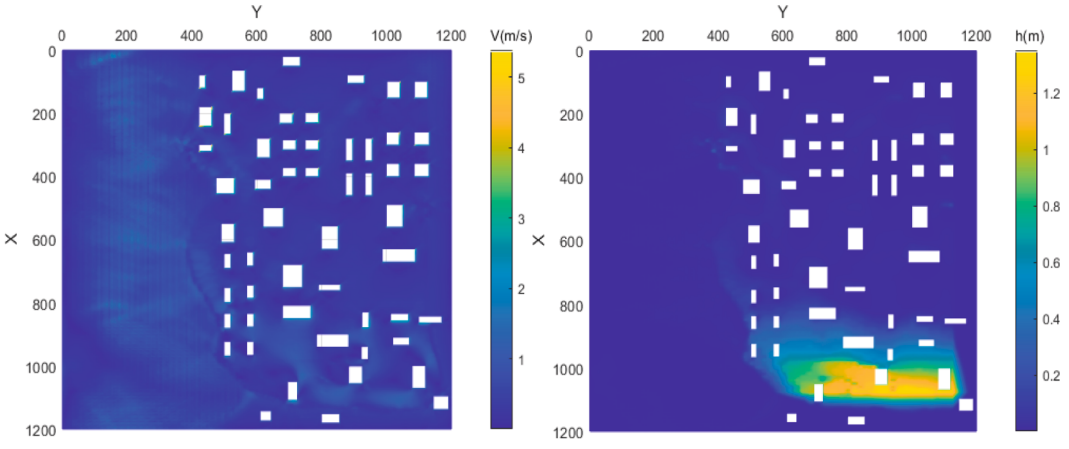
The calculated detour distance matrix, replacing the Euclidean distance, is considered as the heuristic function in simulations. According to the current cell position, a pedestrian will evaluate the distance estimate value $W_T(n)$ of each adjacent accessible cell and move towards to the cell with the minimum $W_T(n)$. If there are multiple cells satisfying the rule, one of them will be randomly selected, and its state will be changed from an accessible cell ($S = 0$) to a cell occupied by a person ($S = 1$). If the estimated distance of all neighboring cells is greater than 900, the pedestrian is trapped and remains in place. The calculation formula for the distance estimate value is

$$W_T(n) = K_T(n) + G_{T_a}(n), \tag{26}$$

in which, $K_T(n)$ is the distance from the current cell to the neighboring cell n at time T, $G_{T_a}(n)$ the detour distance that a pedestrian can predict



(d) T=20 min



(e) T=50 min

Fig. 14. (*continued*).

from the neighboring cell n to the destination cell at time T_a at time T, and $W_T(n)$ the estimated distance in meters, for a person to reach the expected cell from the current cell via the neighboring cell n during the flood at time T.

Meanwhile, a restriction matrix Lim is set to avoid the "infinite loop" caused by local minimum, and to prevent from re-considering some cells that have been entered before during the next search, thereby reducing unnecessary calculations and improving the search efficiency of the algorithm, For a cell (x,y) that a pedestrian has entered, set $Lim(x,y) = 1$, and the pedestrian is prohibited from entering this position in the next move. This restriction will automatically be lifted after 40 movements.

During the flooding, the evacuation route planning can be regarded as a process in which pedestrians, based on the neighborhood conditions, avoid obstacle- risk areas and seek the optimal position for the next movement until they reach the specified safe area where $z(x, y) \geq z_s$.

The A* algorithm also obtains the detour distance matrix through spatial distance superposition. However, it calculates only based on the eight surrounding neighboring cells and does not further compute superposed distances centered on those adjacent cells. Starting from the initial point, the algorithm expands outward to the neighboring nodes, with the search direction determined by the evaluation function defined in Eq. (26).For each neighboring node, the cost value is calculated, and the node with the smallest cost is selected as the next expansion node. By repeating the steps until the endpoint is attained, the complete evacuation path is produced.

Generally speaking, if a pedestrian's response is lagging much behind the flood (the initial ART is a negative number with large absolute value), so that the pedestrian starts to move only after being submerged in a dangerous area, the pedestrian is considered as trapped. The route will be replanned until the trapped state is lifted. If, until the end of the flood evolution, the pedestrian has not reached the safe area, this point is considered to be in a trapped state.

The evaluation of the cumulative evacuation path distance can be performed through the following formula:

$$L = \sum_{i=1}^{m-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}, \tag{27}$$

In which (x_i, y_i, z_i) represents the coordinates of the points that have been reached, here $i = 1, 2, 3, \ldots, m$.

For crowd evacuation, the competitive relationship among pedestrians needs to be considered as well. The first problem to be solved is who should move first in the crowd at a time T. Following ideas of (Cui et al., 2005; Wang, 2008; Chen et al., 2009; Sun et al., 2021), we consider that physical strength, ART, and the degree of panic of pedestrian in the current situation are the key factors reflecting the competitive relationship. Therefore, the following competition rules are proposed:

(i) Sort according to the ART from largest to smallest. The pedestrian

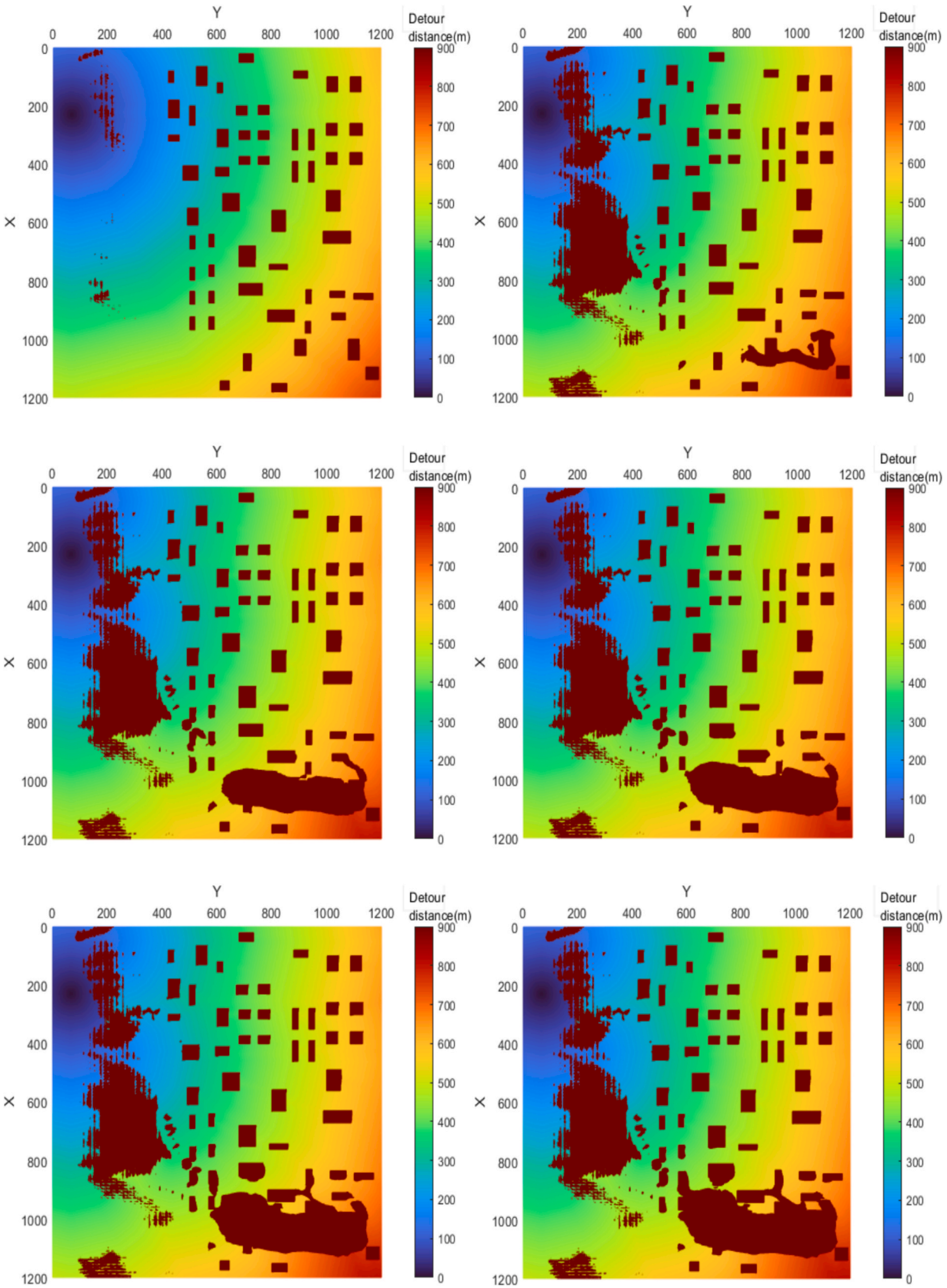


Fig. 15. Detour distance matrices at T = 1, 5, 10, 12, 15, and 20 min. The distance values of obstacle cells are set to 900 m.

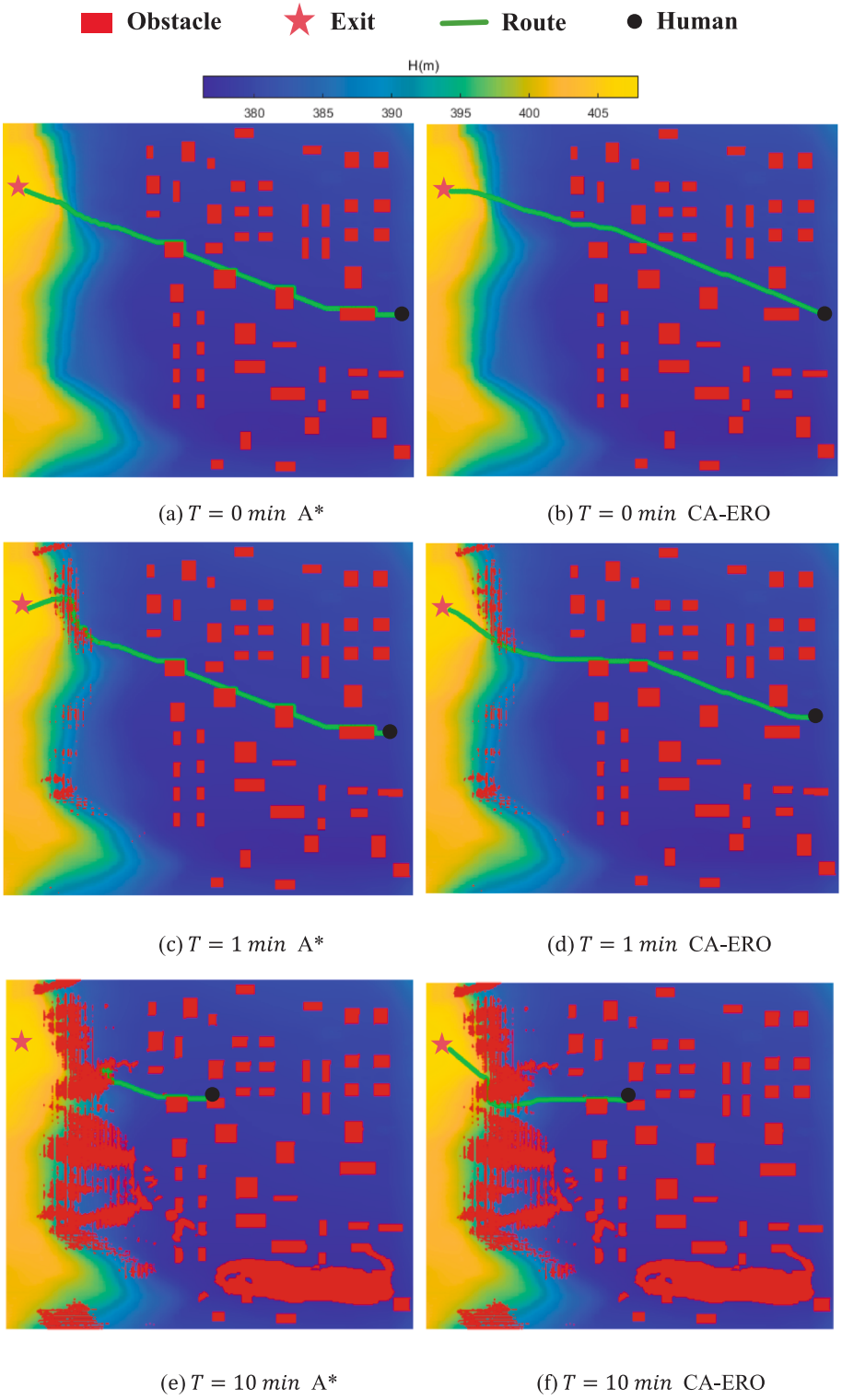


Fig. 16. Evacuation planning route given by Algorithm A* (left) and CA-ERO (right).

with a larger ART moves first.

(ii) When the ART are the same, sort according to physical strength from strongest to weakest. The pedestrian with greater physical strength moves first.

(iii) When both the physical strength and the anticipated response ability to the flood are the same, then consider the degree of panic of people in the flood situation. Sort according to the degree of panic from lowest to highest. The person with the least degree of panic moves first.

(iv) If all three factors are the same, perform a random sorting for movement.

It is important to note that this rule is not intended to cover all possible crowd competition scenarios but instead focus on key factors that primarily affect the competitive dynamics.

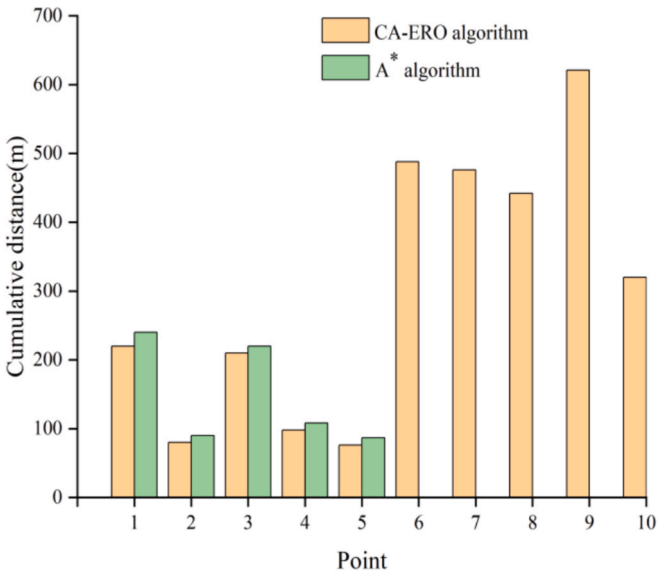


Fig. 17. Cumulative distance for the planed route.

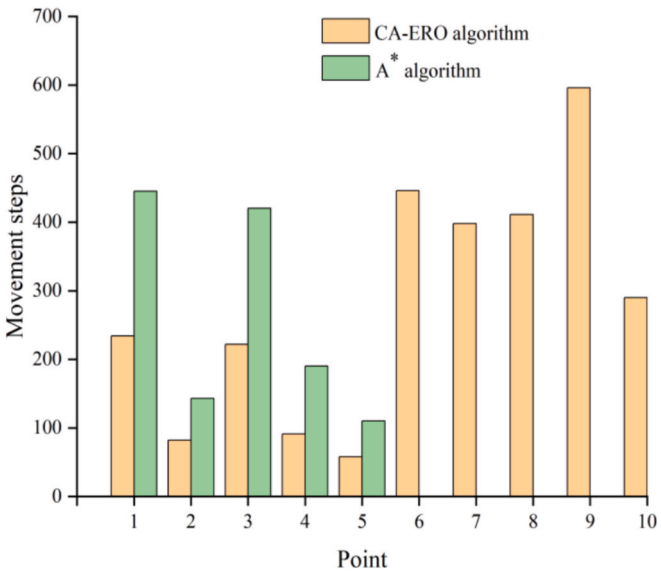


Fig. 19. Number of movements needed for the planned routes.

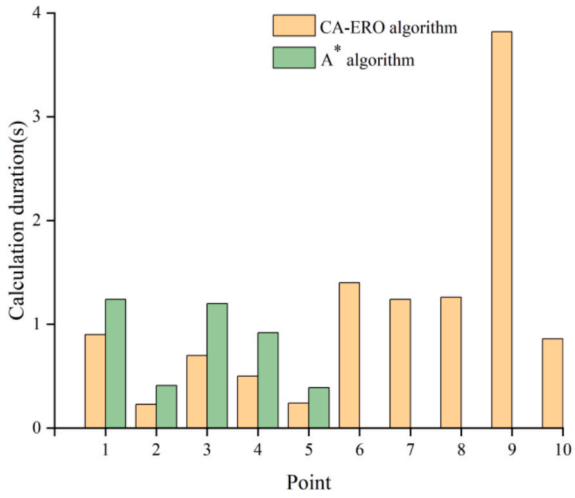


Fig. 18. Computation time needed for successful path planning.

3. Model validation and results

In this section, the correctness of the rainfall term is first verified. Then, the ability of the proposed DBM model to simulate the evolution of urban flood flows is assessed. Finally, we will conduct a case study using an actual suburban area as an example.

3.1. An experiment on rainfall-runoff processes in a simplified urban area

To further assess the performance of the present DBM model in simulating overland flow within urban environments, the rainfall–runoff experiment reported by Cea et al. (2010) is employed for validation. A steel basin is employed to construct the experimental urban catchment, where wooden blocks are used to represent buildings. The basin topography and building layout are illustrated in Fig. 5. The experimental basin has dimensions of 2.5 m × 2.0 m, featuring a central valley that directs flow toward an outlet located along the lower edge. Each building covers a footprint of 20 cm × 30 cm, with vertical exterior walls rising to a height of 20 cm. In the experiment, a rainfall intensity of 300 mm/h was applied for 40 *s*. Since buildings are solid and water cannot flow through them, the areas around the buildings are treated with slip boundary conditions. However, the missing rainfall in this part needs to be compensated by the compensation coefficient $Y = A(\text{Total area})/[A(\text{Total area}) - A(\text{Total area of all building})]$. So the final rainfall intensity is $R \times Y$.

In this simulation, a uniform grid of 500 × 400 cells is employed with a time step of 0.0002 *s*. The Manning roughness coefficient is set to 0.016 across the entire domain, and the reference depth is defined as $h_0 = 0.03m$. For the treatment of dry-wet boundaries, the thin film scheme (TFS) (Oey et al., 2005) is used. During each iteration, the water depth at all computational nodes is compared with the predefined minimum characteristic water depth h_{min}. If the water depth at a node is lower than this value, the node will be marked as a 'dry node'. In this case, the water depth will be forced to h_{min}, and the velocity at this node will be set to zero. In this test, $h_{min} = 0.00003m$. A transmissive boundary is applied at the outlet of the domain, where the macroscopic depth h_b and velocity $\boldsymbol{u}_b = (u_{bn}, v_{bk})$ at boundary are set to be desired values h_i and $\boldsymbol{u}_i = (u_{in}, v_{ik})$, where the subscripts n and k represent the outward positive component and the tangential component, respectively.

At the inlet and along the upper and lower boundaries, reflective boundary conditions are applied, where h_b and v_{bk} are specified as h_i and v_{ik}, respectively, while the normal velocity component u_{bn} is set to zero. The entire simulation runs for 150 *s*.

Fig. 6 compares the simulated discharge obtained from the DBM model with the experimental measurements. The simulation results show good agreement with the experimental data, confirming the accuracy of the DBM model, although the runoff appears to be slightly underestimated after the rainfall ceases. Fig. 7 shows the simulated water depth and flow field when rainfall ceases. Obviously, there is a strong connection between the rainfall-runoff process and the domain's topography. Fig. 8 shows the spatial distribution of the Froude number at $t = 40s$, indicating the occurrence of supercritical flow in the middle and downstream parts of the domain, which also confirms the capability of the DBM in reproducing supercritical flow.

3.2. Laboratory flash-flood experiments over an idealized urban block

The urban flash-flood experiment conducted by Testa et al. (2007) was performed within the framework of the European IMPACT (Investigation of extreMe flood Processes And unCerTainty) program. In this study, a 1:100 scale physical replica of the Toce River valley was constructed at the CESI laboratory in Milan, Italy. To emulate the urban fabric, concrete cubes with 15 cm edges were arranged in a staggered

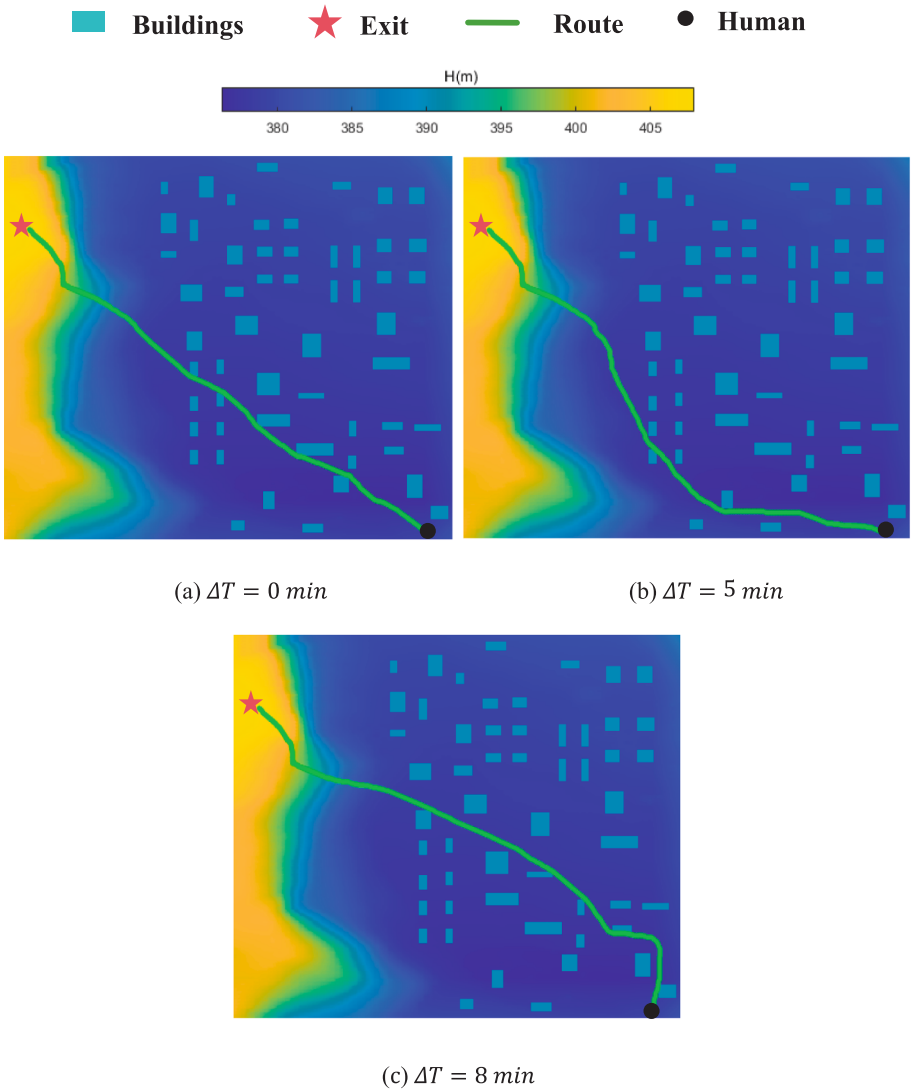


Fig. 20. Evacuation route planned by the CA-ERO algorithm with different initial ART.

configuration, as illustrated in Fig. 9(a).The inflow conditions were generated using a head tank positioned at the upstream (left) boundary, producing the hydrograph presented in Fig. 9(b). This inflow corresponds to the "high-discharge'' scenario described by Testa et al. (2007). Throughout the 60 s test period, water levels were recorded continuously at several instrumented locations, whose positions are indicated in the figure. At the downstream end of the model (right boundary), an open-flow boundary was applied.

Following the recommendation of Testa et al. (2007), the Manning roughness coefficient is prescribed as 0.0162. The numerical simulation employs a uniform mesh divided into a 748 × 640. The time step is set to 0.0001 s, with $h_{min} = 0.0001m$ and $h_0 = 0.5m$. In the computational domain, the treatment method for buildings is the same as described above.

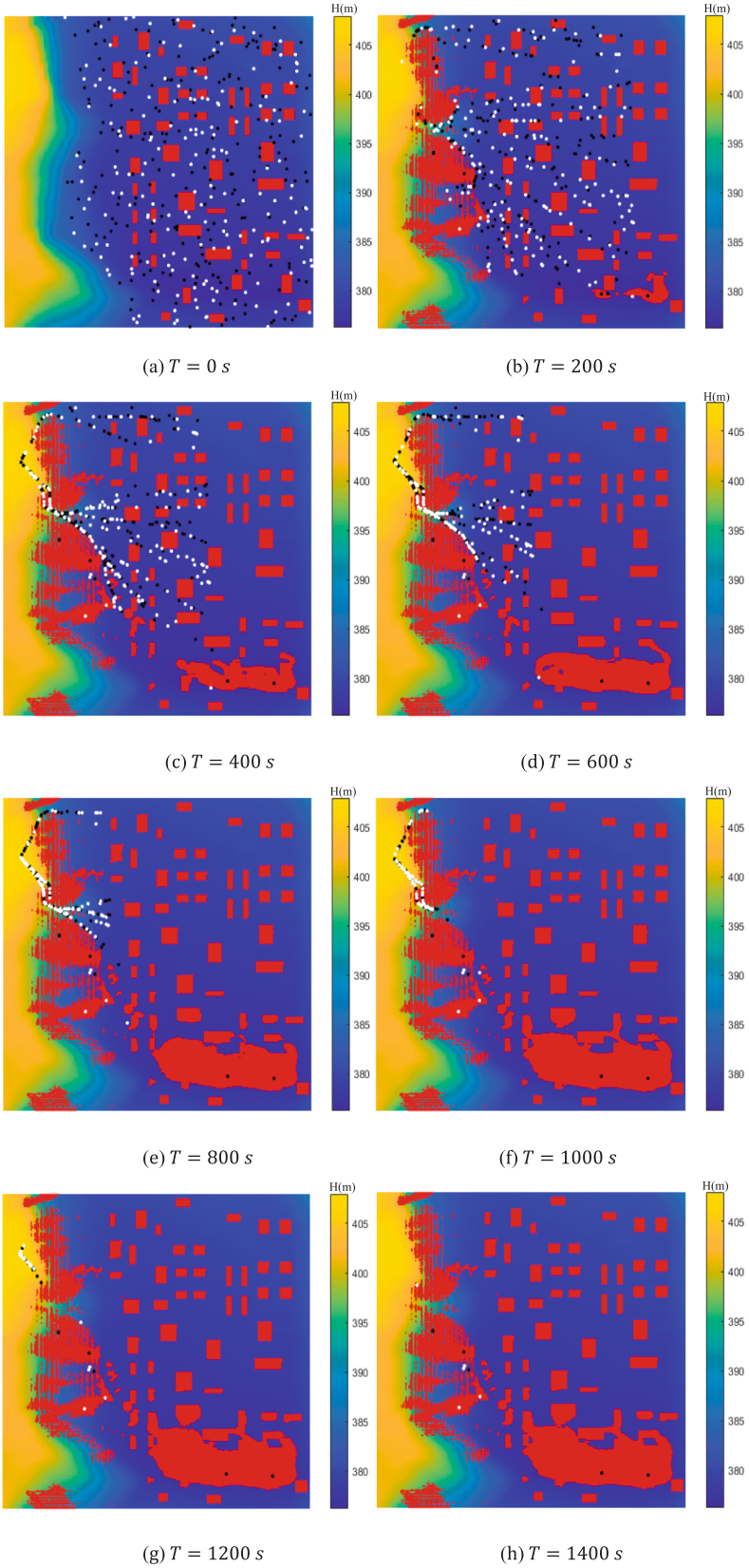
Fig. 10 displays the water-depth fields produced by the DBM at various moments during the simulation. These results demonstrate that the approach effectively shows the transient characteristics of the flash-flood process. Overall, the model succeeds in reproducing key flow phenomena, including hydraulic jumps, recirculation patterns, and wake regions.

Fig. 11 provides an additional validation of the simulation by comparing the computed water depth with the corresponding experimental observations. In general, the numerical predictions reproduce the measured water-level evolution reasonably well and successfully follow the overall temporal variations. Some underestimation is observed at gauges G6 and G8, which is likely associated with the inherent limitations of the shallow-water equations in describing highly transient and turbulent three-dimensional flow features. Overall, this test case confirms that the proposed model is capable of reliably capturing the propagation of flash floods in complex urban settings.

3.3. *Real − world case studies of urban flooding events*

In the following, we will be conducting a case study by taking a suburban region as an example. The DEM plan view and 3D view of this region are shown in Figs. 12 and 13 respectively. Since the resolution of the original DEM data is 5 m, to improve the accuracy in our study, the inverse distance weighted interpolation method is used in GIS to obtain terrain data with a resolution of 0.5 m. However, even with this treatment, it may still have an impact on the results. Therefore, if possible, it is recommended to use high resolution DEM data.

The actual size of region is 600 m × 600 m, with an area of 0.36 km^2. The elevation of the flat ground is approximately 378 m. The elevation range of the hillside is from 390 m to 410 m. The elevation of buildings is uniformly set at 390 m, and the total area of buildings is 0.09896 km^2. According to the average body dimensions of a standing human (Abdelghany et al., 2010), a raster model with a grid size of 0.5 m × 0.5 m is used for modeling in the simulation which divides the space into



(a) $T = 0\ s$ (b) $T = 200\ s$

(c) $T = 400\ s$ (d) $T = 600\ s$

(e) $T = 800\ s$ (f) $T = 1000\ s$

(g) $T = 1200\ s$ (h) $T = 1400\ s$

Fig. 21. Distribution of pedestrians at the time points of 0, 200, 400, 600, 800, 1000, 1200, and 1400 s using the CA-ERO algorithm.

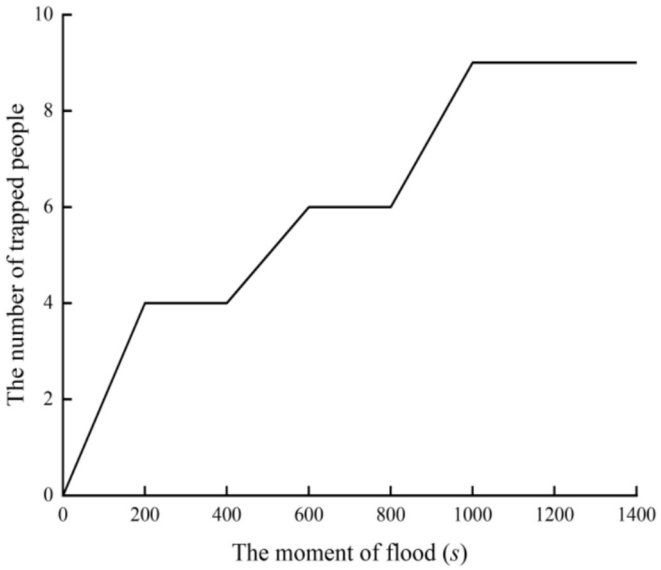


Fig. 22. Number of trapped pedestrians with the time.

Table 3
Sensitivity of evacuation outcomes to ART.

Scenario	Mean trapped	Std trapped	Success rate (%)	Mean evacation time (min)
ART = 0 min	9.18	1.78	97.70	26.65
ART = 5 min	2.27	1.02	99.43	28.48
ART = –5 min	18.10	2.73	95.47	39.90

Table 4
Sensitivity of evacuation outcomes to physical strength.

Scenario	Mean trapped	Std trapped	Success rate (%)	Mean evacation time(s)
M:80–100, F:75–95	9.18	1.78	97.70	26.65
M:60–80, F:55–75	9.6	2.54	97.60	27.53
M:100–120, F:95–115	7.58	1.08	98.10	26.13

1200 × 1200 uniform units. The areas in the terrain where the elevation is greater than 407 m are selected as the safe area. An exit is set in the safe area, with coordinates (229, 68), as shown by the red pentagram in Fig. 13.

3.3.1. Numerical simulation of rainstorm − flood disasters

First, DBM is used to model the flood evolution caused by rainfall. In the hydrodynamic numerical simulation, the rainfall intensity R is taken as 150 mm/h. The region is discretized into 1200 × 1200 uniform cells. The entire model is simulated for 70 min, matching the duration of the rainfall. For the source term, the slope and ground friction are considered, and the Manning roughness coefficient is set to 0.0162. The time step is set to 0.005 s, and the reference length h_0 is set to 1 m. Here, the minimum characteristic water depth h_{min} is taken as 0.0001 m. Transmissive boundary conditions are applied to the upper, lower, left, and right boundaries of the region. During the simulation, the results are exported to the evacuation model every 10 s. These results include the water depth, flow velocity, flow rate, coordinates, and other relevant data for each grid point.

Fig. 14 shows the flood water depth and flow velocity diagrams at $T = 5$ min, 8 min, 10 min, 20 min, and 50 min. The x and y axes in this figure represent the grid. It can be observed that at the beginning of the rainstorm, the water depth changed quite dramatically, in particular the water depth at the low-lying (see the top right corner) gradually increased with the time and then decreased, e.g., at $T = 50$ min. After approximately 20 min, the flood caused by the rainstorm tends to stabilize, and there is no significant change in water depth across the region. The part at the region of x = 480–550 m and y = 380–570 m is a slightly low-lying area in the entire region, making it extremely prone to serious waterlogging, which leads to a rise in water level. Additionally, the flow velocity from the hillside to the flat area is relatively large, which often easily forms debris flows and poses a threat to evacuation. If the flow velocity and water depth diagrams are magnified locally, it can be seen that the simulation results of the multi-speed DBM can accurately capture the water depth and flow patterns around the ground and buildings.

3.3.2. Evacuation route simulation

According to the topographical characteristics, the above-mentioned location (229, 68) with a relatively high terrain was selected as the exit. Then, using the above known initial conditions, the detour distance matrices at each moment were calculated through the cellular automaton algorithm. The detour distance matrices at some moments are shown in Fig. 15. The x and y axes in this figure represent the grid.

The detour distance matrices were obtained after considering the global obstacle distribution and flood evolution. The color represents the magnitude of the detour distance from each point to the nearest exit. However, due to the presence of obstacles, the superposition distances for pedestrians to reach the area behind the obstacles increases. Therefore, in the figures, a phenomenon similar to 'interference' happens behind the obstacles. Moreover, such matrix effectively quantifies the actual connectivity of the terrain and presents the real-world situation.

The flood related process changes over time, and the movement process of people also changes with time. To synchronize these two processes in terms of space and time, we set the time step ratio as 5 cellular automation updates per DBM iteration.

Then, we randomly select a point with grid coordinates (620, 1180) for comparison with the traditional A* algorithm. Fig. 16(a)~(f) show the evacuation planning route of the A* algorithm and the CA-ERO algorithm under flood conditions at the moments of $T = 0$, 1, and 10 min. It can be found from the figures that the route planned by the A* algorithm failed to “see” the obstacles until encountering them.. Since the pedestrian cannot determine the actual connectivity of the terrain, this will increase the detour distance during their evacuation. When the flood evolution process reaches 10 min, the traditional A* algorithm can no longer plan a route to the exit and gets trapped among the obstacles, as shown in Fig. 16(e). In contrast, the CA-ERO algorithm proposed in this study successfully plans the evacuation route throughout the flood evolution and bypasses the obstacle points.

To further demonstrate the advantages of this algorithm, 10 different starting points were randomly selected for evacuation planning. The cumulative distance of the shortest paths planned by the two algorithms is shown in Fig. 17. No data is available at points 6 ~ 10 for the A* algorithm as it failed to plan paths. Moreover, the distance planned by the CA-ERO algorithm is shorter than that of the A* algorithm. As can be seen from Fig. 18, the CA-ERO algorithm needs less calculation time than the A* algorithm. It can also be seen from Fig. 19 that the number of movements of the CA-ERO algorithm is lower than that of the A* algorithm, indicating that the number of algorithm operations is reduced, and the efficiency of planning is higher. Moreover, it can be seen from the simulation that A* algorithm could fail in practical applications.

Next, we take the pedestrian at point (1178, 1131) as an example to examine the impact of the initial ART on evacuation. As shown in Fig. 20, when $\Delta T = 0min$, the distance for the pedestrian to reach the safe area is 746.34 m. When $\Delta T = 5min$, the pedestrian can already be

aware of the danger in advance, so as to maintain a greater distance from the flood and move towards the safe area, the evacuation distance will increase to 806.94 m. When $\Delta T = 8min$, the pedestrian evacuates to the safe area in a completely different direction from before, so the distance increases to 809.15 m. It is worth noting that although the path planned when $\Delta T = 0min$ is the shortest, the pedestrian may walk into the flood, with a potential risk of instability. Although the evacuation distance increases as the initial ART increases, the pedestrian becomes aware of the flood danger earlier and earlier, and takes the measure of detouring in advance, making the path safer. In most cases, people evacuate after the flood occurs rather than before it. Therefore, the evacuation process related to a positive ART still has to face flood obstacles. When the ART is positive, even though the evacuation time and conditions have changed, people still need to bypass these same flood obstacles, resulting in an increase in the evacuation distance.

In reality, usually many people evacuate together. The initial state of the crowd and the flooding jointly determine the evolution of the flood disaster. At the initial moment, 400 people are randomly distributed, with 220 males and 180 females. The matrices for gender, *MF*, physical fitness , *power*, physical fitness consumption rate, *consume*, ART, *PRE*, and panic level, *panic*, are set up with the same size as the above mentioned discrete space. *MF* has only three state values. $MF(x,y) = 1$ indicates a male; $MF(x,y) = 2$ indicates a female; $MF(x,y) = 0$ indicates that there is no one at this position. Random initial physical fitness values are generated. Since males are generally more physically fit than females, the initial physical fitness values at positions where $MF(x,y) = 1$ (males) range from 80 to 100, while at positions where $MF(x,y) = 2$ (females), they range from 75 to 95. The physical fitness consumption rate is a randomly generated physical fitness decay coefficient for a person at the position coordinates (x,y) and at time T. To represent different degrees of physical fitness consumption during a person's movement, the value ranges from 0.98 to 1. The physical fitness value of a person is updated according to Eq. (28) every time they move one step. Meanwhile, at the initial moment, an initial ART, which can be positive, negative, or zero, is given to the persona at a position. And an initial panic level, with values ranging from 0 to 1, is also set. The lower the value, the calmer the person is. Note that when a person moves to a new position at the next moment, they will inherit the gender, ART, and panic level attributes of the previous position.

$$power_{T+1}(x_1,y_1) = power_T(x,y) \times consume_T(x,y) \tag{28}$$

Among them, $power_{T+1}(x_1,y_1)$ is the physical fitness value at the new position (x_1,y_1) at time $T+1$; $power_T(x,y)$ is the physical fitness value at the current moment; $consume_T(x,y)$ is the physical fitness consumption rate at the current moment. If a person is unable to move, $consume_T(x,y)$ will always be equal to 1.

At the beginning of the evacuation, the crowd are randomly distributed as shown in Fig. 21(a). The white dots represent females, the black dots represent males, and the points surrounded by red are the trapped people. Fig. 21 and Fig. 22 show that within less than 200 *s* after the occurrence of rainstorm − induced flooding, four people were already threatened and trapped. The reason is that the speed of the rainstorm induced flood evolution is faster than their evacuation speed. The trapping rate of people is the highest in the early stage because the sudden rainstorm causes rapid waterlogging in low-lying areas. The number of trapped people remains stable in the time intervals of 200 *s* ∼ 400 *s*, 600 *s* ∼ 800 *s*, and 1200 *s* ∼ 1400 *s*. When the flooding process occurs after 600 *s*, it can be clearly observed that the crowd will spontaneously form a forked shaped flow of people (non − linear evacuation situation), which is a common self − organization phenomenon of the flow of people during various disaster evacuations. Crowding mainly occurs in the upper and lower areas near the safety exits. This situation can be regarded as a key concern when formulating flood evacuation plans. Therefore, guidance personnel can be arranged in advance at these key points to provide evacuation guidance, allowing people to pass in an orderly manner or choose alternative routes to detour, so as not to miss the best time for crowd evacuation.

4. Sensitivity analysis of behavioral parameters

In this study, behavioral parameters such as physical fitness, ART, and panic level were mainly derived from literature values or assumptions, which may introduce uncertainties. To evaluate the robustness of the simulation results, a sensitivity analysis was conducted focusing on these two parameters.

(1) ART

The initial anticipated response time reflects the degree of crowd responsiveness to flooding. To examine its influence, three different response levels were analyzed in the context of emergency evacuation under flooding. Following the principle of controlling variables, the procedure was as follows:

① Among the initial population, 220 males and 180 females were included. Each individual was assigned a physical strength value, and the initial states of *MF* and *power* were saved as baseline parameters for all experiments;

② The parameter $PRE^{t=0}(x,y)$ was set to three levels of responsiveness: –5, 0, and 5 min. For each case, the evacuation algorithm was executed 10 times, and the mean number of trapped individuals, the standard deviation of trapped individuals, evacuation success rate, and evacuation time were recorded.

The simulation results (Table 3) show that a smaller ART leads to a significant increase in the number of trapped individuals, whereas a larger ART yields a higher evacuation success rate. This highlights ART as a key behavioral parameter affecting evacuation efficiency. In addition, it can be observed that when ART = 0 min, the mean evacuation time is the shortest. As the initial anticipated response time lags behind the flood onset, the mean evacuation time increases; however, when the response time occurs earlier than the flood, the mean evacuation time also increases. This is mainly because an earlier response prompts individuals to take longer routes for safety, thereby increasing the likelihood of competition with others and even causing congestion at critical nodes.

(2) Physical strength

As described in Section 2.2, when individuals share the same ART and compete for the same position, physical strength is used to determine priority. To investigate the sensitivity of this parameter, three cases were tested:

① The initial population was the same as in the previous setup, with each individual assigned an ART value of zero. The initial states of *MF* and *PRE* were saved as the baseline parameters for all experiments;

② The male and female strength values in $Power^{t=0}(x,y)$ were set as shown in Table 4. Each scenario was simulated 10 times, and the corresponding results were recorded.

Table 4 shows that reducing physical strength increases the number of trapped individuals and prolongs evacuation time, whereas increasing strength produces the opposite effect.

Overall, sensitivity analyses demonstrate that both ART and physical strength influence evacuation outcomes, but ART exerts a greater effect because it directly alters the available time for escape relative to flood propagation. Therefore, the robustness of our model conclusions is confirmed. Nevertheless, future studies should incorporate field survey or behavioral experiment data to calibrate these parameters.

5. Conclusions

In this study, a flood emergency evacuation model was established by combining the geographical cellular automata emergency evacuation model and the multispeed DBM for simulating rainfall flood process. The seamless integration of the structured grid and the numerical results of rainfall runoff hydraulics were successfully achieved. First, the accuracy of the rainfall term was verified using a two-dimensional rainfall case;

then, the capability of the DBM in simulating and evolving urban floods was further verified, and finally, taking the rainstorm-induced flooding in a suburban area of a city as an example, simulation research was carried out. The results show that, compared with the traditional A* algorithm, the proposed model can expand the search field to a global scope, effectively avoid buildings and flooded areas, prevent the phenomenon of detouring along the wall, and has higher solution efficiency and shorter evacuation routes. The simulation can be combined with the dynamic flood process. When the initial ART is $\Delta T = 0$, the optimal evacuation route based on the current water situation can be planned. When the initial ART $\Delta T > 0$, a safer evacuation route can be adjusted and planned based on higher safety considerations. Moreover, considering the impact of floods on evacuation from the perspective of the crowd, the present model can not only simulate the locations and times of congestion but also predict the routes with high traffic volume by using the self-organization phenomenon of the crowd. The nonlinear evacuation situations that were difficult to describe in previous flood emergency evacuation models were also explained. Meanwhile, sensitivity analyses confirm the robustness of our model, highlighting that ART plays a more dominant role than physical strength.

However, as mentioned earlier, some shortcomings remain. There is a lack of data for verifying the rainfall-runoff hydraulics model. In conjunction, there is no measured evacuation result data to test the effectiveness of the evacuation algorithm. In addition, the human behavior parameters in the evacuation are all obtained through literature research or temporarily assumed to be a reasonable value. In the actual application, these parameters can be further calibrated through physical model experiments, investigation of crowd parameters, and an analysis of the actual environment. These are all issues that need to be further addressed in future research.

CRediT authorship contribution statement

Ting Zhang: Writing – original draft, Methodology, Investigation. **Yong Peng:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Jianping Meng:** Writing – review & editing, Supervision, Software, Methodology. **Jianguo Zhou:** Writing – review & editing, Methodology, Investigation. **Xianfei Yin:** Validation, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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